



Punjab University College of Information Technology (PUCIT)
University of the Punjab (PU), Lahore.

Answer Sheet No. _____

Name: _____ BSSE Time: 120 min Marks: 160

Roll No: _____ Section: (Encircle) Morning Afternoon
(mention your PF section)

Signature: _____ Degree: (Encircle) BSE BCS BIT

Programming Fundamentals

Final Exam

INSTRUCTIONS



Do Not Start Until You Are Advised To Do So

1. Do not forget to pray before starting to attempt the paper. It helps. Remember! SOMEONE is always with you (so be relaxed), and HE is also watching you (so be honest). Have Faith and Trust Yourself.
2. This is a close book, close notes exam.
3. No Queries, Understanding the question is part of exam.
4. There are 12 printed pages including this sheet. Check them before leaving.
5. If you feel space for a question is not enough, write in two columns.
6. Electric media (Calculator, Mobile, PDA, Laptop etc) is strictly prohibited in the examination hall. Cheating is highly un-ethical and will cause the cancellation of paper.
7. Remember, time management is critical, so be professional.

B E S T O F L U C K ! ! !

Question 1 [40]	Question 2 [60]	Question 3 [60]	Total [160]	Total [40]

Signature: _____ Student: _____ Teacher: _____

Extra page: use in case of extreme emergency

Dry-run the following codes, assuming no errors.

A.

<pre>double a[3] = {-2.50, 3, 2.6}; double *p = &a[1]; cout << p[+1] << endl; cout << p[0] << endl; cout << p[-1] << endl;</pre>	Rough work	Output
--	------------	--------

B.

<pre>struct Box { int h, w, l; }; int main() { Box b; b.h = 3; b.w = 5; b.l = 2; Box *p = &b; cout << "base is "; cout << p->l * p->w << endl; return 0; }</pre>	Rough work	Output
--	------------	--------

C.

<pre>int a[3][3] = {{1,2,4},{5,3,7},{0,9,6}}; int *p[3]; p[1] = a[2]; p[2] = a[0]; p[0] = &a[1][1]; cout << p[2][2] << endl; cout << p[0][1] << endl; cout << p[0][2] << endl; cout << p[1][2] << endl; cout << p[2][3] << endl;</pre>	Rough work	Output
--	------------	--------

D.

<pre>char m[] = "BSEF16 Finals"; cout << strlen(m) << endl; cout << strchr(m, 'F') << endl; char* p = strrchr(m, 'F'); cout << p-m << endl; strcat(m, " exams"); cout << m << endl; cout << &m[9] << endl;</pre>	Rough work	Output
--	------------	--------

E.

<pre>stringstream ss; ss.str("6376 484 28332 373"); char c; double d; int i; string s; ss >> d >> c >> i; cout << d << " " << c << endl; ss >> c >> c >> s; cout << s << " " << i << endl;</pre>	Rough work	Output
--	------------	--------

Write code/functions for the following descriptions.

1. Complete the following code. You have to compute and display the net (final) price based on quantity, rate, and appropriate discount. Store is offering 5% discount on purchase quantity above 10 units and 12% discount on purchase quantity above 25 units. Hint: price= quantity * rate

```
double quantity, rate;
cout << "enter quantity and rate ";
cin >> quantity >> rate;
```

2. Write code (with #includes) to display printable ASCII chart in the following form using loops.

```
32      33 !   34 "   35 #   36 $   37 %   38 &   39 '
40 (   41 )   42 *   43 +   44 ,   45 -   46 .   47 /
...
120 x 121 y 122 z 123 { 124 | 125 } 126 ~
```

3. Write code to print the sum of elements of both diagonal of a square matrix (2DArray) of order **size**.
OR: Write recursive function to return the sum of elements of its main diagonal.

- Write down the contents (including new lines, tabs, etc) of the file generated by the following code.

<pre>#include <iostream> #include <fstream> using namespace std; int main() { int a = 837; int b = 26; char c = 32; // ascii of space character float d = 0.025; char * p = &c;</pre>	<pre>ofstream s; s.open("file.txt"); s << "PUCIT, Punjab University" << endl; s << "Old Campus, Lahore" << endl; s << endl; s << a << '\t' << b << *p << d << endl; return 0; }</pre>
---	---

What's wrong/missing in the above code, if any.

- Write down function to return the sum of a two rational numbers (of rational type), when declaration of structure **rational** is given as:

```
struct rational
{
    int p, q;
};
```

- Declare a structure of type course consists of C strings for course name (30 characters) and course code (8 characters) and an integer for credit hours. Later write code to save contents an array of N such structures into a binary file named **courses.dat**.

Write code/functions for the following descriptions.

- A. Consider data arrays to stores only positive numbers. A programmer has written several functions to manipulate such data arrays by placing data values at contiguous indices from 0 onwards and place an end of data marker after them. Rest of allocated space in array is garbage and never considered as data. He has taken that end of data marker as -1 a negative value which never be appeared as data. A lot of work is remaining and deadline is approaching, and fortunately you are hired to code some functions to manipulate such data. Here you have to provide code to join (append) data in array named **nd** to array named **data**, assuming **data** array have necessary and sufficient space.
- ```
void join(int data[], int nd[])
```

- B. In the scenario of previous part A, if **data** array may not have necessary and sufficient space and physical sizes (allocated space) of data arrays is known in main logic, you have to write code of an overloaded **join** function with appropriate prototype. It is required to reuse code by calling **join** function of part A from within it.

- C. Given below is structure of a text file. Its first line contains only an integer representing number of cities stored in it. Second line is headings and third line make these headings underlined. You have to read data

of cities from file (cities.txt) and store names, area and population of cities having area upto 500 in appropriate array(s).

2

| City          | Area | Population | Logitude  | Lattitude |
|---------------|------|------------|-----------|-----------|
| Lahore        | 1772 | 6318745    | 74.329376 | 31.582045 |
| Rahimyar Khan | 833  | 832645     | 69.457104 | 32.733351 |

- D. Consider the scenario given in previous part C, this time challenge is to update population of a city. You have to write code to get city name and its new population from user and using minimum file reading and writing, search city name in it and if found update it population with new value.

- E. Given code is used to create a 2D dynamic array of `int` data type of dimensions D1 and D2. You have to write code used to create a 3D dynamic array of dimensions D1, D2 and D3 of float data type. Also write code to deallocate memory allocated to given 2D array.

```

int **array = new int*[D1];
for(int j=0; j<D1; j++){
 array[j] = new int[D2];
}

```

5. Write implementation of the following function related to type complex (complex numbers).

|                                                                                                                                                                                                                                          |                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| <pre> complex conjugate(const complex &amp;); double magnitude(const complex &amp;); complex add(const complex &amp;, const complex &amp;); bool equal(const complex &amp;, const complex &amp;); void show(const complex &amp;); </pre> | <pre> struct complex {     float real;     float imag; }; </pre> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|





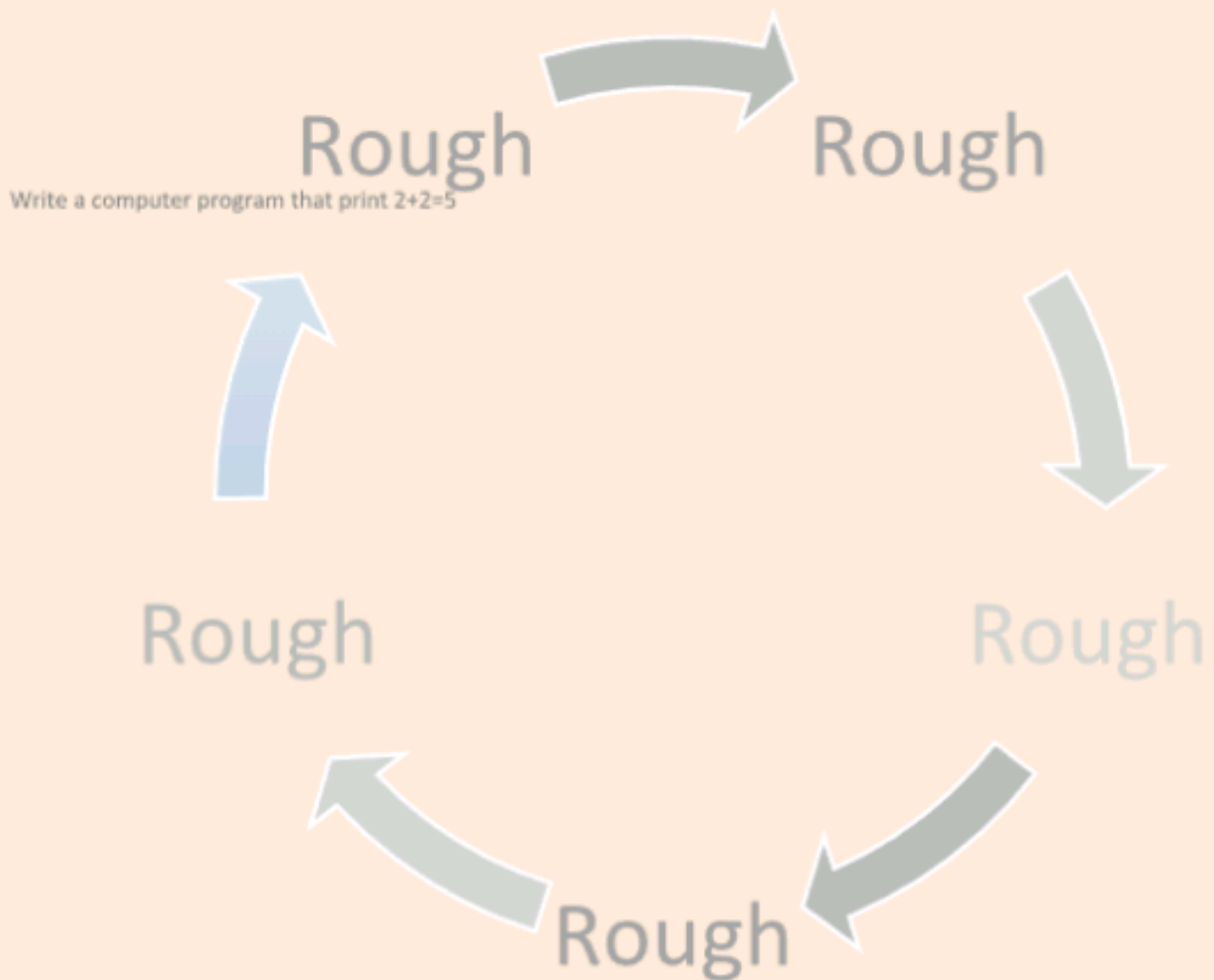
**Replacement questions, you may replace any question of paper but with 50% credit.**

- A. Write a recursive function called *sumover* that has one argument *n* which is an unsigned integer. The function returns a double value which is the sum of reciprocals of the first *n* positive integers. For example, *sumover*(1) returns 1.0 and *sumover*(2) returns 1.5 as it is  $1/1+1/2$ .
- B. Consider a four dimensional array contains the percentage marks for students, first dimension is for program no (0,1,2 and 3), second dimension is for semester no (0,1,2,3,4,5,6 and 7), third dimension is for subject no (0,1,2,3,4 and 5), fourth dimension is for student no (0,1,2,3,4,...,199). You have to, write a function to compute and display the subject's (passed as parameter) average.
- ```
void subavg(float a[4][8][6][200], int subjno)
```

ROLLNO : _____ SIGN : _____

HONOUR CODE _____

In the movie, Rise of the Planet of the Apes, a chimpanzee is solving which puzzle?

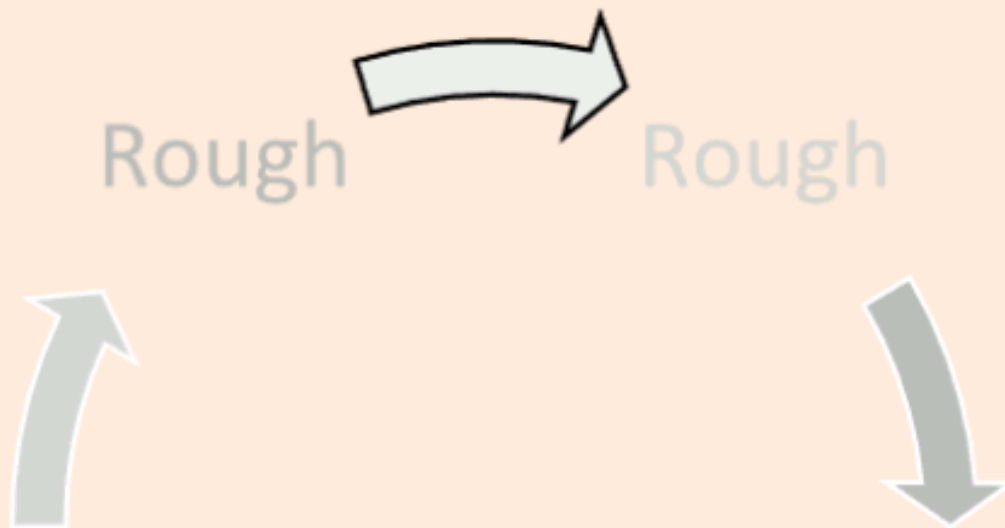


Why scientists always think differently? E.g. when bacteria under experiments were dissolved, why not Fleming abuse the blue-green mold who is responsible of the contamination.

ROLLNO : _____ SIGN : _____

HONOUR CODE _____

Which type of program is printed on the marriage invitations?



In the movie, 3 Idiots, are the below ranked graduated idiots failed to enter in the market?



Why PUCIT students study only for marks?